TIDJANI BACHIR

Full-Stack Developer

Blida, Algeria

+213 6 98 03 33 14

bachirtidjani769@gmail.com

GitHub: github.com/Tidjani1Bachir

Portfolio: personnel-website-dusky.vercel.app

SUMMARY

Full-Stack Developer specializing in secure MERN applications and cross-platform desktop tools. I build production-ready software with real security depth — JWT auth, bcrypt, 4-layer file validation, and NoSQL injection protection — not just tutorial-level implementations. Currently completing a Master's in Software Engineering at the University of Saad Dahleb, Blida.

PROJECTS

Full-Stack E-Commerce Platform — MERN Stack:

Architecture & Security: Built with JWT Bearer token authentication, bcrypt password hashing, 4-layer file upload validation (extension, MIME type, path traversal, malware scanning), and NoSQL injection protection via express-mongo-sanitize with explicit type casting across all controllers

- **State Management:** RTK Query tag-based cache invalidation for real-time UI sync on all admin CRUD operations; hybrid localStorage/MongoDB favorites for guest-to-user data migration
- **Payment & Dashboard:** Integrated PayPal SDK for PCI-compliant payments; admin dashboard with live ApexCharts analytics and CSV/PNG/SVG exportable reports
- **Deployment:** Decoupled architecture — React on Vercel, Node.js/Express API on Render, MongoDB Atlas cloud database with automated seeding scripts

Product Manager — Web + Desktop App:

- **Live App:** https://product-manager-chi-eosin.vercel.app
- **Desktop (.exe):** https://github.com/Tidjani1Bachir/product_manager/releases/download/v2.0-desktop/product-manager_0.1.0_x64-setup.exe
- **Desktop (.msi):** https://github.com/Tidjani1Bachir/product_manager/releases/download/v2.0-desktop/product-manager_0.1.0_x64_en-US.msi
- **Stack:** React, TypeScript, Tauri, Node.js/Express with full CRUD, drag-and-drop image upload, and PDF export
- **Deployment:** Dual-version architecture — web app on Vercel (serverless) and native desktop installer (.exe / .msi) via Tauri from a single GitHub repository
- **Cloud:** Integrated Turso (cloud SQLite), Cloudinary (image storage), and RESTful Express API for a zero-infrastructure-cost cloud-native stack
- **Testing:** 39/39 passing tests — Vitest + React Testing Library (unit), Playwright E2E across Chrome, Firefox & Safari, and node:test for REST API validation
- **TypeScript Migration:** Migrated full codebase from JavaScript to TypeScript with strict typing, shared interfaces, and a centralized API service layer

EDUCATION

Master of Science in Software Engineering (In Progress) 2024 – Present

University of Saad Dahleb – Blida, Algeria

- Relevant coursework: Software Architecture, Advanced Development Methodologies, Project Design

Bachelor of Science in Computer Systems 2020 – 2024

University of Saad Dahleb – Blida, Algeria

- Final Year Project: Lawyer-client management platform using Firebase (backend) and Android Studio / Java (mobile)
- Features: real-time database sync, user authentication, appointment & case management

Scientific Baccalaureate — Bien (Above Average) 2021

Houari Mahfoud High School – Blida, Algeria

- Specialization: Experimental Sciences

TECHNICAL SKILLS

Frontend: React, Next.js 14, TypeScript, Tailwind CSS, Redux Toolkit / RTK Query, Headless UI, Framer Motion, Syncfusion EJ2

Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcrypt, Multer, Rate Limiting, File Upload Validation, express-mongo-sanitize

Database: MongoDB Atlas, MongoDB Compass, Turso (cloud SQLite), Firebase

DevOps: Vercel, Render, GitHub Pages, GitHub Actions, Tauri (desktop), Cloudinary

Testing: Vitest, React Testing Library, Playwright (E2E), node:test, Postman (API testing)

Tools: Git & GitHub, Figma, ApexCharts, PayPal SDK

OS & Scripting: Linux (Debian), Bash scripting, WSL, VirtualBox/VMware, Vim, Ollama (local AI)

Other: UML Diagrams (Use Case, Class, Activity, State Machine, Sequence), Agile, Clean Architecture, MVC Pattern

Languages: Arabic (Native), English (Intermediate), French (Intermediate)